GLP–Graph Behaviour Under Hyperparameter Variation

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September 2026

Abstract

This paper presents a systematic study of the GLP–Graph (GLPG) architecture under extensive hyperparameter variation, revealing that GLPG behaves not merely as a compact language model but as a scalable and interpretable cognitive substrate. Where the companion architecture paper defines GLPG's structural foundations— deterministic graph routing, learned lemma geometry, and dual generative modes— the present work investigates how these mechanisms evolve across changes in model scale, embedding dimension, paging geometry, context vector design, sampling regimes, and training depth.

Across several controlled ablations, GLPG exhibits a consistent developmental trajectory: proto-semantic attractors in underscaled models, emergent semantic manifolds in mid-scale configurations, and canon compression in large models. These transitions are not statistical artefacts but structural phase changes driven by the architecture itself. GLPG maintains collapse-resistance, stable world-model formation, and biome-specific stylistic adaptation even in minimal configurations, demonstrating that coherent language behaviour can arise from a low-dimensional dynamical system rather than from large-scale attention mechanisms.

The results identify an optimal manifold width, an effective embedding blend coefficient, a minimal functional context vector, and a robust baseline architecture. More importantly, they show that GLPG's behaviour scales cognitively rather than numerically: increasing capacity deepens attractor basins and strengthens canon pressure, while decreasing capacity increases plasticity and creative drift without destroying coherence. These findings position GLPG as a modular, interpretable, and domain-faithful substrate for hybrid intelligent systems within the Triadic Cosmos ecosystem.

Significance Statement

The GLP–Graph (GLPG) architecture is not presented here as a specialised language model, but as a minimal and interpretable cognitive substrate within the broader *Triadic Cosmos* ecosystem. The ecosystem aims to unify conceptual theory, computational architectures, and empirical demonstrations into a coherent framework for studying intelligence and AGI. Within this context, GLPG serves as a concrete, executable instance of a low-dimensional dynamical system capable of producing coherent linguistic behaviour without relying on large-scale attention mechanisms or statistical correlation.

The hyperparameter study reported in this paper shows that GLPG exhibits structured cognitive dynamics across scale: proto-semantic attractors in small models, emergent semantic manifolds in mid-scale configurations, and canon compression in larger models. These transitions mirror developmental phases observed in biological and artificial cognitive systems, demonstrating that language-like behaviour can arise from a compact, modular substrate governed by deterministic routing, learned lemma geometry, and stable attractor basins.

Crucially, GLPG maintains coherence, stylistic adaptation, and world-model formation even in minimal configurations, revealing that its generative capabilities stem from architectural principles rather than parameter count. This positions GLPG as a foundational component of the Triadic Cosmos program: a scalable, interpretable, and structurally grounded substrate that illustrates how intelligence can emerge from unified cognitive dynamics rather than from statistical scale alone.

1 Introduction

The GLP–Graph (GLPG) architecture, introduced in the companion paper [1], establishes a hybrid generative substrate that unifies deterministic graph routing with learned lemma geometry inside a single monolithic MLP. GLPG was designed not as a specialised language model but as a minimal, modular, and interpretable cognitive system capable of producing coherent linguistic behaviour through structural and semantic dynamics rather than through large-scale statistical correlation. Its dual generative modes, paged graph traversal and pure MLP sampling, reveal an architecture that combines explicit structural determinism with emergent semantic manifolds.

The present paper investigates how this substrate behaves under systematic variation of its hyperparameters. Whereas the architecture paper defines the mechanisms that make GLPG possible, the current work examines the conditions under which those mechanisms become expressive, stable, brittle, or transformative. We conduct extensive ablation studies across model scale, hidden-layer geometry, output embedding dimension, paging configuration, sampling regimes, context vector design, activation functions, recursion mechanisms, and training depth. These experiments are performed on multiple canonical datasets spanning different narrative biomes, allowing us to observe how GLPG's internal dynamics adapt across domains and across scales.

Our findings show that GLPG exhibits structured cognitive behaviour across hyperparameter regimes. Underscaled models produce proto-semantic attractors and highly plastic narrative drift; mid-scale models develop coherent semantic manifolds and stable biome adaptation; large models compress their internal worlds into rigid canonical structures with deep attractor basins. These transitions are not numerical artefacts but architectural phase changes: GLPG behaves as a dynamical cognitive system whose manifold geometry, attractor structure, and narrative coherence evolve predictably with capacity.

This behaviour is significant within the broader *Triadic Cosmos* ecosystem, which aims to unify conceptual theory, computational substrates, and empirical demonstrations into a coherent framework for studying intelligence and AGI. GLPG provides a concrete example of a minimal cognitive substrate: a low-dimensional, interpretable system that generates coherent language through deterministic routing, semantic injection, and stable attractor dynamics. By showing how GLPG deforms, stabilises, and reorganises its internal manifold under hyperparameter variation, this paper demonstrates that coherent linguistic behaviour can emerge from compact cognitive principles rather than from large-scale attention mechanisms.

The goal of this work is therefore twofold: to identify an optimal baseline configuration for GLPG, and to show that its generative behaviour arises from cognitive dynamics intrinsic to the architecture. The results provide both a practical foundation for future GLPG-based systems and a conceptual bridge between minimal cognitive substrates and scalable hybrid intelligent systems within the Triadic Cosmos program.

2 Experimental Setup

All experiments in this paper are performed on a single canonical dataset: *Alice in Wonderland, Retold in Words of One Syllable* (Gutenberg). This dataset is intentionally chosen for its unique combination of grammatical simplicity, semantic clarity, and narrative completeness. It is a full book, not a fragment, yet small enough to permit extensive ablation studies while remaining structurally rich. Its constrained vocabulary and regular syntactic cadence make it an ideal substrate for observing how GLPG's internal dynamics respond to hyperparameter variation.

2.1 Dataset Characteristics

The one-syllable Alice corpus contains:

- 97 KB of raw text,
- 222 KB of canonical grammar–lemma tokens,
- 331 sequences after preprocessing,
- 17,179 unique samples,
- a target vocabulary of 1,660 grammar–lemma pairs.

At 20,000 training epochs, the model processes approximately one million learned token transitions, meaning that each unique sample is encountered roughly 57 times. This repetition is sufficient to induce stable world-model formation without overfitting, allowing us to observe GLPG's developmental phases under controlled conditions.

2.2 Default Model Configuration

Unless otherwise specified, experiments begin from a standard baseline architecture:

$$592 \rightarrow 1024 \rightarrow 512 \rightarrow 512 \rightarrow 512 \rightarrow 64,$$

a monolithic MLP with four hidden layers and a 64-dimensional lemma output embedding. Paging is configured with a maximum of 512 pages, though the dataset naturally occupies 204 pages when restricted to 16 inputs per page. This provides a balanced manifold width: narrow enough for deterministic routing, yet broad enough for semantic modulation.

2.3 Generative Modes and Evaluation

Each configuration is evaluated in both generative regimes supported by GLPG:

- **Paged graph traversal**, which restricts candidate transitions via deterministic last-lemma routing, and
- **Pure MLP mode**, which evaluates the full vocabulary and exposes the learned semantic manifold.

For every tested hyperparameter setting, the model generates ten independent stories of twenty lines each in both modes. These outputs are analysed for:

- syntactic stability,
- semantic coherence,
- narrative flow,
- attractor formation and collapse behaviour,
- biome-specific stylistic adaptation,
- recursion robustness,
- and overall cognitive profile.

This setup provides a controlled environment in which the effects of model scale, embedding dimension, paging geometry, sampling parameters, context vector design, and training depth can be observed directly and compared systematically. The simplicity of the Alice corpus ensures that emergent behaviour arises from the architecture itself rather than from dataset complexity, making it an ideal benchmark for studying GLPG as a cognitive substrate.

3 Training Epochs

Training depth plays a central role in shaping GLPG's internal dynamics. Because the architecture learns through repeated traversal of a fixed canonical dataset, the number of epochs directly determines how strongly structural cadence, semantic motifs, and attractor basins become internalised. This section examines four training regimes on the Alice one-syllable corpus, using the default model architecture and paging configuration.

3.1 Experimental Results

Table 1 summarises the behaviour of the default Alice model under different epoch counts. All other hyperparameters remain fixed.

Size (MB)	Train (s)	Paged Gen (s)	MLP Gen (s)	Scenario & Result
21.0	49	38	86	**3K epochs**: underfit. Grammar cadence emerges, but semantic fields are minimal; attractors are shallow and narrative flow is fragmented.
21.0	174	50	110	**10K epochs**: mid-regime. Semantic motifs begin to form; clause chains stabilise; paging becomes more reliable.
21.0	296	37	127	**20K epochs (default)**: broad but chaotic. Strong semantic emergence, rich variation, and stable paging. Pure MLP mode shows high flux but remains coherent.
21.0	478	48	194	**30K epochs**: overfit. Attractor basins deepen, creative drift decreases, and narrative becomes rigid with strong canon pressure.

Table 1: Training epoch ablation for the default Alice model.

3.2 Dataset-Derived Epoch Heuristics

The Alice corpus contains 331 sequences and 17,179 unique samples. At 20,000 epochs, the model processes roughly one million learned token transitions, meaning each unique sample is seen approximately 57 times. This density is sufficient to induce stable world-model formation without collapse.

Two empirical heuristics emerge:

- **Epochs per canonical dataset size:**

$$\frac{20{,}000}{222\text{ KB}} \approx 90 \text{ epochs per KB.}$$

  This ratio consistently yields strong semantic emergence on compact canonical corpora, where only grammar–lemma tokens contribute to manifold formation.

- **Epochs per sequence:**

$$\frac{20{,}000}{331} \approx 60 \text{ epochs per sequence.}$$

This ensures repeated traversal of each structural unit, allowing lemma embeddings to stabilise and attractor basins to form.

3.3 Cognitive Dynamics Across Training Depth

GLPG exhibits a clear developmental progression as training depth increases:

- **Underfit (3K)**: structure without meaning; weak attractors; fragmented narrative flow.
- **Mid-regime (10K)**: semantic fields begin to form; paging stabilises; clause chains lengthen.
- **Optimal regime (20K)**: rich semantic emergence, balanced variation, strong attractor basins, and expressive pure MLP behaviour.
- **Overfit (30K)**: excessive attractor depth; reduced plasticity; rigid canon; diminished creative drift.

Unlike transformer models, GLPG does not collapse under extended training. Instead, excessive training compresses the semantic manifold, reducing variability while preserving coherence. For the Alice corpus, 20,000 epochs represent the optimal balance between semantic richness, structural stability, and narrative flexibility.

4 Hidden Layer Sizes

The GLPG architecture employs a monolithic MLP with four hidden layers. In the default configuration, the first hidden layer is significantly larger (1024 units) than the subsequent three layers (512 units each). This asymmetry was initially motivated by the intuition that the first layer must absorb the full context vector and project it into a sufficiently expressive semantic manifold. However, the ablation results below show that increasing the first layer alone provides limited benefit, and that balanced layer sizes often yield more stable and expressive behaviour.

4.1 Experimental Results

Table 2 summarises the behaviour of the Alice model under different hidden-layer configurations. All experiments use the same dataset, paging geometry, and sampling parameters; only the hidden-layer widths are varied.

4.2 Interpretation

Several clear patterns emerge from these ablations:

- **Increasing only the first layer yields limited benefit.** The 2048-unit first layer slightly enriches semantic variation, but the improvement is small relative to the computational cost. GLPG's semantic manifold does not depend strongly on a single wide layer.
- **Balanced layers outperform asymmetric configurations.** When all four layers are set to the same width (e.g. 768), the model becomes more stable but also less expressive. Balanced layers reduce flux and stylistic variation, producing a narrower manifold.
- **Small layers remain surprisingly functional.** Even extremely small configurations (32/16/16/16 or 16/8/8/8) retain structural coherence. This confirms that GLPG's routing and grammar filtering provide a strong backbone independent of MLP capacity.
- **Semantic expressiveness scales with total manifold width.** Larger layers deepen attractor basins and improve biome adaptation, while smaller layers increase creative drift but reduce coherence.

Size (MB)	Train (s)	Paged Gen (s)	MLP Gen (s)	Scenario & Result
33.9	337	56	144	**2048 first layer**: small improvement. Slightly richer semantic manifold, but diminishing returns relative to compute cost.
27.7	308	44	137	**All layers = 768**: weaker than hybrid configurations. Stable but less expressive; reduced stylistic variation.
9.7	272	64	132	**128 first layer**: stable but compressed. Semantic fields form, but manifold width is limited.
9.5	259	51	203	**64 other layers**: stable but fragmented. Paging remains coherent, but semantic continuity weakens.
2.8	272	36	100	**128 first, 64 others**: functional but less expressive. Canon remains intact; semantic drift increases.
1.9	242	49	154	**32 first, 16 others**: stable but minimal. Produces coherent syntax but very shallow semantic attractors.
1.8	226	33	281	**16 first, 8 others**: stable with minimal expression. Telegraphic behaviour; pure MLP mode becomes noisy.
1.7	230	27	58	**2 first, 1 others**: stable without expression. Structure survives, semantics collapse; resembles DMLG micro-model behaviour.

Table 2: Hidden-layer size ablation for the default Alice model.

4.3 Cognitive Dynamics

These results highlight an important property of GLPG: the MLP acts primarily as a *semantic router* rather than as a high-capacity transformer-like encoder. Even very small MLPs can sustain coherent generation because the deterministic graph provides structural cadence and routing constraints. Larger MLPs enrich the semantic manifold but do not fundamentally change the system's behaviour.

The optimal configuration for the Alice corpus lies between the extremes: moderately wide layers (512–1024) provide sufficient semantic expressiveness without inducing excessive canon rigidity or computational cost. This balance supports stable attractor formation, coherent narrative flow, and controlled stylistic variation.

5 Output Dimension and Learnable Target Embeddings

GLPG uses learned lemma embeddings as the semantic anchors of its manifold. The dimensionality of these embeddings determines how much semantic structure the model can store in its output space. Unlike transformer architectures, GLPG does not rely on attention layers for semantic organisation; instead, the learned grammar–lemma pair embeddings encode a large portion of the world-model. Varying the output dimension therefore directly affects the depth, breadth, and stability of the semantic attractor landscape.

This section examines output dimensions ranging from a single scalar to 1024-dimensional embeddings, revealing how GLPG transitions from purely graph-driven behaviour to fully emergent semantic modelling.

5.1 Experimental Results

Table 3 summarises the behaviour of the Alice model under different output embedding sizes. All other hyperparameters remain fixed.

Size (MB)	Train (s)	Paged Gen (s)	MLP Gen (s)	Scenario & Result
19.7	259	41	–	**1 dimension**: stable but graph-dependent. Pure MLP mode non-functional; semantic manifold collapses to a line.
19.7	255	42	–	**1 dimension, 64 pages**: still stable with graph routing; paging provides all structure.
19.6	291	86	–	**1 dimension, 16 pages**: minimal behaviour; paging dominates; semantics remain extremely shallow.
19.6	321	337	–	**1 dimension, 4 pages**: almost pure MLP mode. Graph constraints weaken; model becomes unstable and noisy.
19.7	295	37	322	**4 dimensions**: shallow canon, good syntaxis. Semantic clusters begin to form but remain weak.
20.0	256	52	258	**16 dimensions**: diffuse canon. Semantic fields broaden but lose coherence; attractors remain shallow.
22.3	383	50	164	**128 dimensions**: sweet spot. Strong semantic emergence; pure MLP mode matches paged mode; stable attractor basins.
24.9	384	45	171	**256 dimensions**: larger canon. Deeper semantic clusters; pure MLP mode fully functional; slightly reduced variation.
40.5	664	147	594	**1024 dimensions**: oversized. Diminishing returns; manifold becomes over-compressed; compute cost increases sharply.

Table 3: Output embedding dimension ablation for the default Alice model.

5.2 Interpretation

Several clear patterns emerge:

- **1-dimensional embeddings rely entirely on the graph.** GLPG remains structurally coherent because paging and grammar rules provide deterministic routing, but semantic emergence is impossible. Pure MLP mode collapses.

- **Small dimensions (4–16) produce shallow semantic manifolds.** Syntax stabilises, but semantic clusters remain weak and diffuse. Canon formation is inconsistent.
- **128 dimensions represent the optimal balance.** This configuration yields:
  - strong semantic coherence,
  - stable attractor basins,
  - rich biome adaptation,
  - pure MLP mode nearly identical to paged mode,
  - manageable compute cost.
- **256 dimensions deepen the manifold but reduce variation.** Canon becomes stronger; creative drift decreases; attractors become more inertial.
- **1024 dimensions overshoot the manifold capacity.** The model becomes slower, more rigid, and less expressive. Semantic compression reduces stylistic variation.

5.3 Cognitive Dynamics

The output embedding dimension determines the *semantic bandwidth* of GLPG. Larger dimensions allow the model to encode richer world geometry directly in the grammar–lemma embeddings. This explains the remarkable observation that GLPG can function with a single scalar output when paging is active: the graph provides all structural constraints, and the MLP merely selects transitions.

As the dimension increases, the model transitions from:

- **graph-only cognition** (1–4 dims),
- **shallow semantic emergence** (16 dims),
- **full semantic manifolds** (128–256 dims),
- **semantic over-compression** (1024 dims).

The 128-dimensional configuration consistently provides the richest and most balanced cognitive behaviour: expressive yet stable, varied yet coherent, emergent yet structurally grounded. This dimension allows pure MLP mode to model the world as effectively as paged mode, demonstrating that GLPG's learned embeddings contain a large portion of the canonical domain.

5.4 Embedding Alpha

GLPG updates its grammar–lemma embeddings using a fixed blending rule that mixes the stored embedding with the predicted embedding at each transition. The blend coefficient α determines the rate of semantic adaptation: small values preserve long-term stability, while larger values induce higher flux and faster drift in the semantic manifold. Because GLPG relies on stable attractor basins for coherent world-model formation, the choice of α has a direct impact on narrative behaviour, canon width, and MLP expressiveness.

Table 4 summarises the behaviour of the default Alice model under four alpha settings, including the standard configuration $\alpha = 0.001$.

Size (MB)	Train (s)	Paged Gen (s)	MLP Gen (s)	Scenario & Result
21.0	363	61	182	$\alpha = 0$: rigid and narrow canon. Embeddings become static; semantic drift disappears; pure MLP mode weakens significantly.
21.0	296	37	127	**Default** $\alpha = 0.001$: balanced regime. Stable attractor basins, coherent semantics, and expressive but controlled MLP behaviour.
21.0	295	44	200	$\alpha = 0.002$: richer world, moderate flux. Semantic fields expand; MLP mode becomes more expressive; slight increase in noise.
21.0	337	37	250	$\alpha = 0.01$: strong blending, high flux. Semantic manifold becomes highly dynamic; attractor basins destabilise; narrative becomes noisy.

Table 4: Embedding blend coefficient ablation for the default Alice model.

Interpretation

The blend coefficient controls the *semantic plasticity* of GLPG:

- $\alpha = 0$ freezes the semantic manifold. The model behaves almost like a pure graph system: structurally coherent but semantically narrow, with reduced variation and weaker MLP behaviour.
- **Default** $\alpha = 0.001$ provides the most stable behaviour. Attractor basins deepen gradually, semantic motifs stabilise, and pure MLP mode remains expressive without collapsing. This regime consistently yields coherent world-models across datasets.
- $\alpha = 0.002$ increases semantic richness. The manifold becomes broader and more varied, but also slightly noisier. MLP mode benefits from the increased plasticity.
- $\alpha = 0.01$ induces excessive flux. Embeddings drift too quickly, semantic clusters dissolve, and narrative coherence decreases. Paging remains stable, but pure MLP mode becomes noisy and unstable.

Cognitive Dynamics

Alpha determines how quickly GLPG "rewires" its semantic manifold during training. Small values allow the model to accumulate stable semantic structure over many epochs, mirroring slow consolidation in biological cognitive systems. Large values induce rapid plasticity, which increases variation but destabilises long-term coherence.

Across all datasets tested, the range $\alpha \approx 0.001$–0.002 consistently produces the strongest cognitive behaviour: expressive yet stable, varied yet coherent, and resistant to collapse. This regime allows GLPG to refine its semantic geometry without erasing previously learned attractors, making it the preferred setting for both paged and pure MLP generation.

6 Amount of Hidden Layers

Depth in GLPG determines how many nonlinear refinements the semantic projection undergoes before cosine scoring. Unlike transformer architectures, additional layers do not introduce hierarchical attention patterns; instead, they modulate the flux dynamics of the semantic manifold and determine how expressive or rigid the attractor landscape becomes. This section evaluates configurations ranging from one to eight hidden layers, under different embedding blend coefficients α.

6.1 Experimental Results

Table 5 presents the results. We separate the configurations into three groups: (A) aggressive blending ($\alpha = 0.01$), (B) default blending ($\alpha = 0.001$), and (C) mild variations ($\alpha = 0.002$ and $\alpha = 0.005$).

6.2 Interpretation

Three key patterns emerge:

- **Aggressive blending ($\alpha = 0.01$) amplifies flux.** Shallow networks collapse into telegraphic behaviour; deeper networks become rigid and over-compressed. Pure MLP mode becomes fast but noisy.
- **Default blending ($\alpha = 0.001$) is stable across depths.** The default 4-layer model provides balanced expressiveness and stability. Depth increases semantic richness but also reduces creative drift.
- **Mild blending ($\alpha = 0.002$–0.005) improves 5-layer models.** These settings produce the richest and most coherent behaviour: attractors deepen without collapsing, semantic fields broaden, and narrative flow becomes fluent.

6.3 Cognitive Dynamics

Depth and blending interact strongly:

- **Depth increases manifold complexity.** More layers refine the semantic projection, deepening attractor basins.
- **Blending controls manifold plasticity.** Small α stabilises semantics; large α destabilises them.
- **Optimal cognitive regime: 5 layers + $\alpha = 0.002$–0.005.** This combination yields:
  - rich semantic emergence,
  - stable attractor geometry,
  - fluent narrative flow,
  - balanced creative drift,
  - controlled flux.

Overall, GLPG behaves as a dynamical system with an optimal depth. Too few layers limit expressiveness; too many layers reduce variation and induce rigidity. For the Alice corpus, five hidden layers with mild blending produce the strongest cognitive behaviour.

Size (MB)	Train (s)	Paged Gen (s)	MLP Gen (s)	Scenario & Result
A. Aggressive blending ($\alpha = 0.01$)				
9.4	203	94	–	**1 layer**: story-focused but only graph mode works; semantic manifold too shallow for pure MLP mode.
14.9	259	37	192	**2 layers**: fully functional; pure MLP slow but coherent; attractors begin to form.
17.9	251	47	255	**3 layers**: strict, telegraphic canon; limited semantic variation.
24.0	375	40	145	**5 layers**: pure MLP much faster; broad and stable manifold; reduced noise.
27.0	419	40	84	**6 layers**: pure MLP faster; small improvement in coherence.
30.0	486	27	85	**7 layers**: consistent but telegraphic; manifold becomes rigid.
33.0	718	33	79	**8 layers**: oversized; reduced expressiveness; no real benefits.
B. Default blending ($\alpha = 0.001$)				
21.0	296	37	127	**Default 4 layers**: balanced regime. Stable attractor basins, coherent semantics, expressive but controlled MLP behaviour.
24.0	381	40	120	**5 layers**: less flux, less interesting than $\alpha = 0.01$; stable but narrow manifold.
27.0	407	52	125	**6 layers**: rich but chaotic manifold; increased variation.
30.0	454	37	78	**7 layers**: fast, expressive, deep manifold; reduced creative drift.
33.0	468	30	62	**8 layers**: fast but oversized; semantic compression reduces variation.
C. Mild blending variations ($\alpha = 0.002$ **and** $\alpha = 0.005$)				
24.0	380	30	84	**5 layers,** $\alpha = 0.002$: more flux, improved expressiveness; balanced variation and stable attractors.
24.0	375	40	145	**5 layers,** $\alpha = 0.005$: sweet spot. Rich, stable, broad, fluent semantic behaviour.

Table 5: Hidden-layer count ablation for the Alice model under different embedding blend coefficients.

7 Forward Layer Recursion

The previous section showed that increasing depth in a naïve feedforward MLP improves semantic expressiveness but eventually leads to rigidity and over-compression. GLPG, however, is not limited to simple depth scaling: the architecture supports recursive refinement blocks that transform the MLP into a dynamic substrate with internal iterative behaviour. These recursion mechanisms allow the model to perform multiple semantic refinement steps within a single forward pass, effectively increasing functional depth without increasing parameter count.

This section evaluates several recursion strategies, including residual skips, gated refinement, self-feedback loops, and hybrid combinations. The results demonstrate that recursion fundamentally changes GLPG's cognitive dynamics and leads to a new baseline architecture.

7.1 Experimental Results

Table 6 summarises the behaviour of the Alice model under different recursion mechanisms. All experiments use hidden sizes of 768 unless otherwise noted and a default of four semantic hidden layers.

Size (MB)	Train (s)	Paged Gen (s)	MLP Gen (s)	Scenario & Result
21.3	403	57	190	**1024 units, 2 layers, repeat layer 2 (3×)**: narrow manifold; attractor hardening; limited semantic variation.
27.7	375	39	168	**R-skip**: clear upgrade. Stable, expressive manifold; improved semantic continuity.
34.4	586	63	114	**R-gate**: larger upgrade. Iterative refinement with selective retention; stronger attractor basins.
27.7	309	39	113	**Self-feedback**: coherent but modest improvement; lacks selective gating.
27.7	339	37	98	**Refinement loop**: demonstrates need for gating and residuals; unstable without selective control.
34.4	491	41	116	**Hybrid (R-gate + R-skip)**: best upgrade. Stable, less chaotic, consistent manifold; improved narrative flow; fast generation.
34.4	934	44	122	**Super hybrid**: excessive recursion; slower and worse; semantic over-compression.

Table 6: Recursion mechanism ablation for the Alice model.

7.2 Interpretation

Several conclusions emerge:

- **Simple repetition (repeat layer 2)** increases functional depth but hardens attractors and reduces variation.
- **Residual skips (R-skip)** improve stability and semantic continuity by preserving previous states.
- **Gated recursion (R-gate)** introduces selective refinement, allowing the model to retain or overwrite semantic features.
- **Self-feedback and ungated loops** refine semantics but lack control, leading to instability.
- **Hybrid recursion** (R-gate + R-skip) is the strongest configuration: stable, expressive, less chaotic, and consistent across datasets.
- **Excessive recursion (super hybrid)** leads to semantic over-compression and reduced variation.

7.3 The Hybrid Recursion Block

The hybrid block combines gated residual recursion with an additional refinement step. It performs three gated refinement iterations followed by a residual projection. This structure provides both stability and expressiveness.

The forward pass is:

- initial projection into semantic space,
- three gated residual refinement iterations,
- one ungated residual enhancement,
- final projection to output embeddings.

Formally:

$$h \leftarrow h + g \cdot (u - h)$$

where:

$$g = \sigma(W_{\text{gate}} h), \qquad u = \text{silu}(W_{\text{refine}} h).$$

This mechanism allows the model to selectively update semantic features while preserving stable attractor geometry.

7.4 Cognitive Dynamics

Recursion transforms GLPG from a static feedforward network into a dynamic system:

- **R-skip** preserves long-term semantic structure.
- **R-gate** enables selective semantic retention.
- **Hybrid recursion** creates a controlled iterative refinement process.

The result is a deeper, more stable semantic manifold with:

- stronger attractor basins,
- reduced chaotic drift,
- improved biome adaptation,
- more coherent narrative flow,
- faster pure MLP generation.

7.5 New Baseline Architecture

The hybrid recursion block, combined with five hidden layers (four semantic layers + one gating layer), forms the new baseline GLPG architecture. This configuration consistently outperforms naïve depth scaling and provides the best balance between:

- semantic richness,
- structural stability,
- compute efficiency,

- narrative coherence,
- attractor robustness.

Recursion is therefore not an optimisation but a **qualitative architectural improvement** that shifts GLPG into a higher cognitive regime.

8 Activation Functions

Activation functions determine the energy profile of GLPG's semantic manifold. Because the architecture relies on stable attractor basins, controlled flux, and smooth semantic refinement, the choice of activation has a direct impact on narrative coherence, stylistic variation, and manifold stability. This section evaluates several activation functions under identical conditions, revealing that SiLU provides the strongest qualitative behaviour for the computational cost.

8.1 Experimental Results

Table 7 summarises the behaviour of the Alice model under different activation functions. All experiments use the same baseline configuration.

Size (MB)	Train (s)	Paged Gen (s)	MLP Gen (s)	Scenario & Result
21.0	274	39	132	**Hardtanh**: cold, low-energy manifold; clamped dynamics; low variation.
21.0	275	38	127	**Tanh**: coherent but low variation; limited semantic emergence.
21.0	281	41	135	**ReLU**: fragmentary canon; half-clamped behaviour; blocky syntax.
21.0	266	51	125	**Leaky ReLU**: slightly better than ReLU; still limited semantic depth.
21.0	296	37	127	**SiLU**: baseline activation.
21.0	455	48	234	**GELU**: significantly slower; less interesting than SiLU; diffuse manifold.
21.0	295	47	266	**Cosine-Gate Activation**: slower and more diffuse; weaker attractor basins.
21.0	576	50	153	**SiLU + CGA**: different but not better; increased diffusion; reduced coherence.
21.0	503	40	151	**SiLU Cos-gate**: more diffuse than pure SiLU; less stable.

Table 7: Activation function ablation for the Alice model.

8.2 Interpretation

The results show a clear hierarchy:

- **Hardtanh and Tanh** produce low-energy manifolds. They clamp semantic variation and suppress emergence. Syntax remains coherent but narrative flow is flat.
- **ReLU and Leaky ReLU** introduce discontinuities. Canon becomes fragmentary; syntax becomes blocky; attractor basins become unstable due to half-clamping.

- **GELU** is smooth but computationally expensive. Despite its popularity in transformers, GELU performs worse in GLPG: slower training, slower generation, and more diffuse semantics.
- **Cosine-based activations** (CGA, SiLU+gate) increase diffusion. They weaken attractor basins and reduce coherence.
- **SiLU is the strongest performer.** It provides:
  - smooth gradients,
  - stable attractor geometry,
  - rich semantic emergence,
  - coherent narrative flow,
  - excellent compute efficiency.

8.3 Cognitive Dynamics

Activation functions determine the "energy regime" of GLPG's semantic manifold:

- **Clamped activations** (hardtanh, tanh) suppress flux and reduce semantic emergence.
- **Discontinuous activations** (ReLU variants) destabilise attractor basins and fragment syntax.
- **Smooth but heavy activations** (GELU) diffuse semantics and slow computation.
- **SiLU provides optimal manifold dynamics.** Its smooth, monotonic curve supports stable refinement, controlled flux, and expressive semantic fields. It matches GLPG's recursive substrate exceptionally well.

8.4 Conclusion

Activation functions have a logical and measurable impact on GLPG's generative behaviour. Across all tested configurations, **SiLU consistently provides the best qualitative output for the computational cost**. It supports stable attractor formation, rich semantic emergence, and coherent narrative flow, making it the preferred activation for the GLPG architecture.

9 Inputs per Page

Paging determines how many canonical transitions are grouped into a single structural unit of the GLPG graph. Each page contains a fixed number of input transitions, and the model selects the next lemma by evaluating only the transitions within the current page. This mechanism provides deterministic routing and structural cadence. Varying the number of inputs per page therefore changes the balance between determinism and semantic emergence.

9.1 Experimental Results

Table 8 summarises the behaviour of the Alice model under two extreme paging configurations: fully deterministic paging (1 input per page) and near-MLP-only behaviour (128 inputs per page). The default setting is 16 inputs per page.

Size (MB)	Train (s)	Paged Gen (s)	MLP Gen (s)	Scenario & Result
21.1	286	32	177	**1 input per page (1120 pages)**: fully deterministic; canon becomes narrow; semantic emergence limited.
20.9	274	57	175	**128 inputs per page (176 pages)**: closer to pure MLP mode; slower generation; more semantic emergence but reduced structural discipline.

Table 8: Paging geometry ablation for the Alice model.

9.2 Interpretation

The number of inputs per page determines the strength of GLPG's structural constraints:

- **1 input per page** produces a fully deterministic system. The graph dominates; syntax is stable; canon becomes narrow; semantic emergence is minimal. This configuration resembles a rule-based generator with limited creativity.
- **128 inputs per page** weakens graph determinism. The model behaves closer to pure MLP mode; semantic emergence increases but structural discipline decreases; generation becomes slower.
- **Default 16 inputs per page** provides the best balance. It preserves deterministic routing while allowing meaningful semantic variation. Attractor basins remain stable, and narrative flow remains coherent.
- **8 inputs per page** may be preferable for larger datasets. It increases structural resolution without collapsing emergence, offering finer-grained routing for more complex corpora.

9.3 Conclusion

Paging geometry has a clear and predictable impact on GLPG's behaviour. **Full determinism is possible with 1 input per page**, but this suppresses semantic emergence. **High-input pages behave like pure MLP mode**, increasing variation but reducing structural coherence. For the Alice corpus, **16 inputs per page is the optimal setting**, and **8 inputs per page** is a promising configuration for larger datasets.

Paging therefore acts as a structural dial that controls the balance between deterministic routing and emergent semantics.

10 Sampling Parameters

All sampling experiments in this section use the same baseline model: the 20,000-epoch Alice configuration described earlier. Because GLPG is a dynamical system, sampling parameters directly modulate the flux of the semantic manifold. Low top-k values suppress variation and deepen attractor basins, while higher values broaden the manifold and increase narrative energy. Boosting further amplifies flux by reweighting the top candidates.

This section evaluates top-k values from 2 to 20, with and without boosting, and compares several boosting schedules.

10.1 Experimental Results

Table 9 summarises the behaviour of the model under different sampling configurations.

Paged Gen (s)	Scenario & Result
54	**top-k = 2, no boosting**: zero flux; low energy; minimal variation.
64	**top-k = 3, no boosting**: narrow canon; strong attractors; rigid flow.
48	**top-k = 5, no boosting**: low-flux regime; coherent but conservative.
42	**top-k = 5, boosting**: increased energy; broader variation.
38	**top-k = 8, no boosting**: mid-flux regime; broader canon; stable.
38	**top-k = 8, boosting**: increased energy; *sweet spot*.
42	**top-k = 10, no boosting**: high flux; broad canon; borderline stability.
41	**top-k = 10, boosting**: very high flux; near collapse; unstable.
43	**top-k = 15, no boosting**: flux too high; diffuse canon.
38	**top-k = 15, boosting**: excessive flux; fuzzy structure.
53	**top-k = 20, no boosting**: diffuse, noisy, low quality.
36	**top-k = 20, boosting**: over-flux; over-broad manifold; collapse.
42	**top-k = 8, boosting 6-5-4-3-2**: improved boosting; stable.
40	**top-k = 8, boosting 5-4-3-2**: better than 6-5-4-3-2.
39	**top-k = 8, boosting 4-3-3-2-2**: slightly more elegant; best schedule.

Table 9: Sampling parameter ablation for the 20K-epoch Alice model.

10.2 Interpretation

Several clear patterns emerge:

- **Top-1 is non-functional.** The model collapses into deterministic routing with no semantic emergence.
- **Top-2 and top-3 produce extremely low flux.** Canon becomes narrow; attractors deepen; variation collapses.
- **Top-5 is the low-flux regime.** Coherent but conservative; limited stylistic variation.
- **Top-8 is the optimal manifold width.** Balanced flux, coherent canon, rich variation, stable attractors.
- **Top-10 approaches instability.** Flux becomes high; manifold broadens; borderline collapse.
- **Top-15 and top-20 exceed the stability limit.** Canon becomes diffuse; noise increases; structure degrades.
- **Boosting increases flux.** Mild boosting improves variation; strong boosting destabilises the manifold.

10.3 Boosting Schedules

Boosting reweights the top candidates, increasing narrative energy. The results show:

- Fibonacci boosting (13-8-5-3-2) is too aggressive for top-20.
- Mild boosting improves top-8 significantly.
- The best schedule is **4-3-3-2-2**, producing elegant and stable variation.

10.4 Cognitive Dynamics

Sampling parameters modulate the flux of GLPG's semantic manifold:

- low top-k $\to$ deep attractors, narrow canon, low flux,
- mid top-k $\to$ balanced flux, coherent emergence,
- high top-k $\to$ excessive flux, diffuse manifold, collapse.

Boosting acts as an energy amplifier: beneficial in moderation, destructive when too strong.

10.5 Conclusion

GLPG exhibits an optimal manifold width around **top-8**. With tuned boosting (especially **4-3-3-2-2**), this regime provides the richest and most coherent generative behaviour. Larger top-k values introduce chaos, while smaller values suppress emergence.

Sampling therefore acts as a direct control over the dynamical state of GLPG, confirming that the model behaves as a **nonlinear dynamical system** rather than a static language model.

11 Context Input Vector Tuning

The input vector determines how GLPG perceives its local semantic and structural state. Unlike transformer architectures, GLPG does not rely on large contextual windows or attention mechanisms; instead, it uses a compact, hand-engineered context vector that encodes structural anchors, semantic continuity, and minimal narrative memory. This section evaluates several input-vector designs and introduces a new optimized baseline based on the results of all previous ablations.

11.1 Experimental Results

Table 10 summarizes the behaviour of the model under different input-vector configurations. All experiments use the new baseline architecture:

- 20K epochs (1M token transitions),
- 128-dimensional output embeddings,
- 768-unit hidden layers (592–768–768–768–768–128),
- hybrid recursive forward pass with 768-unit gating layer,
- top-8 sampling with 4-3-3-2-2 boosting.

11.2 Interpretation

The results reveal several key insights:

- **Current sentence (CSL)** is the structural anchor. It provides positional and grammatical grounding but lacks semantic continuity.
- **Token History provides semantic continuity.** It encodes lemma-driven temporal dynamics with decay, producing canon-consistent semantic drift.
- **Narrative memory contributes minimally.** Large memory vectors add noise; small memory (8 dims) adds stability without overfitting.

Size (MB)	Train (s)	Paged Gen (s)	MLP Gen (s)	Vector Size	Scenario & Result
35.9	552	47	146	592	**Old baseline**: large legacy vector; functional but inefficient.
30.8	622	140	385	9	**Sequence + line**: extremely small; valid phrases but no meaning.
31.8	648	94	155	128	**Narrative memory (128)**: fast generation; very low semantic value.
33.3	659	178	–	300	**Compressed context**: minor structure; pure MLP mode extremely slow.
32.0	490	45	113	155	**Current sentence**: main contributor; strong semantic anchor.
31.0	560	34	90	35	**Reduced current**: still functional; compact structural signal.
31.1	466	31	83	44	**CSL**: reduced current + sequence + line; stable without memory.
31.6	634	38	106	108	**Narrative memory (64) + CSL**: fully functional; less noise.
31.3	643	37	106	48	**Token history (48, 0.95)**: requires CSL; semantic continuity.
31.3	449	38	105	76	**Token history (32, 0.95) + CSL**: history behaviour; stable.
31.9	533	36	101	108	**Token history (64, 0.95) + CSL**: bleed; less selective.
31.6	560	33	82	92	**Token history (48, 0.95) + CSL**: *sweet spot*.
32.1	600	28	87	140	**Narrative memory (48) + history (48) + CSL**: no added value.
31.7	586	33	95	100	**Narrative memory (8) + history (48) + CSL**: new optimized vector.

Table 10: Context input vector ablation for the optimized GLPG baseline.

- **Compressed context is ineffective.** It contributes little structure and slows pure MLP mode.
- **Sequence + line number alone is surprisingly functional.** The model can generate syntactically valid text without any semantic history, demonstrating the strength of GLPG's structural substrate.

11.3 CSL and Token History: Complementary Roles

CSL (Current + Sequence + Line) provides:

- structural anchoring,
- positional grounding,
- grammatical cadence,
- attractor stability.

Token History provides:

- semantic continuity,
- temporal drift,
- lemma-driven context,
- decay-controlled memory,
- canon-consistent transitions.

Individually:

- CSL without history $\rightarrow$ coherent syntax but no semantic flow.
- Token History without CSL $\rightarrow$ semantic drift but attractor collapse.

Only the combination **CSL + Token History** yields:

- coherent micro-scenes,
- stable narrative flow,
- controlled semantic drift,
- low noise,
- strong attractor geometry.

11.4 The Token History Algorithm

Token History is a lemma-driven, decay-controlled temporal memory:

$$h_t = \alpha h_{t-1} + (1 - \alpha) e_t$$

with first positional rotation:

$$h_t = \text{rotate}(h_t)$$

and selective update:

$$e_t \in \{\text{lemma embeddings}\}, \quad \text{only if grammar token} \in \text{history grammar tokens}$$

and blacklist filtering:

$$\text{if lemma} \in \text{history lemma blacklist} \Rightarrow \text{skip.}$$

This produces a low-noise, canon-consistent semantic drift vector.

11.5 New Baseline Input Vector

The optimized input vector is:

$$35 + 48 + 8 + 9 = 100 \text{ floats}$$

Six times smaller than the legacy 592-float vector.

- **Reduced current (35)**: lemma pos, grammar pos, last grammar, last lemma, comma flag.
- **Token History (48)**: optimal width and decay.
- **Narrative memory (8)**: minimal stabilizer.
- **Sequence + line (9)**: structural grounding.

11.6 Conclusion

The optimized 100-dimensional input vector provides:

- stronger semantic continuity,
- cleaner structural anchoring,
- lower noise,
- faster generation,
- more coherent narrative flow,
- better attractor stability.

GLPG remains significantly less powerful than transformer architectures, but it does not aim to be one. Instead, it provides a compact, interpretable cognitive substrate with emergent behaviour driven by structural cadence and semantic drift.

12 Final New Baseline Model and Minimum Viable Model

All previous ablations converge on a new optimal GLPG architecture for the Alice dataset. This section presents the final baseline model and contrasts it with the minimum viable model (MVM). Both models share the same cognitive substrate and dynamical principles, but differ in capacity, stability, and emergent narrative coherence.

12.1 Architectures

The final baseline MLP uses wide hidden layers and a large output embedding, combined with the hybrid recursive forward pass and the new input vector:

$$\textbf{Baseline: } 100 \to 1024 \to 1024 \to 1024 \to 1024 \to 128 \quad \text{with 1024-unit gate}$$

The minimum viable model uses narrow hidden layers and a minimal output embedding:

$$\textbf{MVM: } 100 \to 64 \to 64 \to 64 \to 64 \to 16 \quad \text{with 64-unit gate}$$

Both models use:

- the optimized 100-dimensional input vector,
- hybrid recursive forward pass,
- 20K epochs (1M transitions),
- 128 lemma-output dimension (baseline) or 16 (MVM),
- top-8 sampling with 4-3-3-2-2 boosting.

12.2 Experimental Results

Size (MB)	Train (s)	Paged Gen (s)	MLP Gen (s)	Scenario & Result
53.4	616	35	121	**Final new baseline**: stable attractors; rich semantic emergence; coherent microfiction; strong biome adaptation; fast generation.
1.4	309	35	105	**Minimum viable model**: proto-narrative sequences; semantic-syntactic attractor dynamics; expressive but shallow; surprisingly coherent given size.

Table 11: Comparison between the final baseline model and the minimum viable model.

12.3 Classification of the Baseline Model

The baseline GLPG model is best described as:

- a small recurrent language model with a custom semantic-memory vector,
- a hybrid RNN-like architecture with structural anchoring and topic-event continuity,
- a non-transformer microfiction generator with strong local coherence.

Closest analogues include GPT-Neo 125M and GPT-2 Medium (345M), though GLPG uses a fundamentally different cognitive substrate and a narrow canon specialization.

12.4 Classification of the Minimum Viable Model

The minimum viable GLPG model behaves as:

- a low-dimensional semantic–syntactic attractor network,
- a minimal recurrent cognitive substrate producing proto-narrative sequences.

Functionally, its closest equivalents are:

- a very small Elman-style RNN with learned embeddings,
- a minimal GRU-like recurrent core without gates,
- a micro-scale Hopfield-like attractor system generating token rhythms,
- a proto-language model comparable in scale to early 1990s RNN text generators.

None of these match GLPG's behaviour exactly, but they approximate its regime: a tiny recurrent dynamical system with semantic drift, syntactic rhythm, and proto-narrative attractor loops.

12.5 Scaling Comparison

Transformers scale gradually:

$$\text{GPT-2 mini (124M)} \;\rightarrow\; \text{GPT-2 large-scale variants (up to 5B)} \quad (\approx 38\times)$$

Same architecture, same dynamics, more capacity $\rightarrow$ *gradual improvement*.
GLPG scales categorically:

$$\text{GLPG-MVM (1.4 MB)} \;\rightarrow\; \text{GLPG-baseline (53 MB)} \quad (\approx 38\times)$$

Same cognitive substrate, same dynamical principles, but:

- attractor basins stabilize,
- semantic manifolds deepen,
- narrative coherence emerges,
- biome adaptation becomes robust.

This is not a gradual improvement — it is a **phase transition**.

12.6 Conclusion

The final baseline model represents the fully realized GLPG cognitive architecture: stable attractors, coherent narrative flow, rich semantic manifolds, and fast generation. The minimum viable model represents the underlying cognitive kernel: a proto-narrative attractor system capable of semantic drift and syntactic rhythm.

Both models share the same "brain," but differ in capacity. GLPG therefore exhibits **categorical scaling**: increasing capacity does not merely improve quality, but unlocks new cognitive behaviours.

13 Other Gutenberg Datasets

To verify that the new baseline architecture is not Alice-specific, we trained the same model on three additional Gutenberg datasets. The results show that GLPG's cognitive behaviour is reproducible across corpora: downscaled models retain the same dynamical regime at lower capacity, while upscaled models exhibit the same attractor-stabilisation and narrative coherence observed in Alice.

13.1 The strange case of Dr. Jekyll and Mr. Hyde

139 KB, 341 KB canonical, 26,706 unique samples, 201 sequences.

- Baseline (1024 hidden, 128 output): 30K epochs, 3,986,140 transitions, 2950 s, 58 MB model.
- Downscaling (64 hidden, 8 output): 30K epochs, 3,946,415 transitions, 829 s, 2.9 MB model.

This test confirms that GLPG performs **cognitive style-compression** even at 3 MB. The small model is not a weaker version of the large model, but a different cognitive phase: narrow attractor basins, strong stylistic cadence, and monomodal narrative loops.

13.2 The Outline of Science, Vol. 1

600 KB, 1423 KB canonical, 106,360 unique samples, 809 sequences.

- Baseline (1024 hidden, 128 output): 100K epochs, 13,171,314 transitions, 8135 s, 68 MB model.
- Downscaling (256 hidden, 64 output): 30K epochs, 3,925,607 transitions, 1701 s, 15 MB model.

Both models exhibit the same dynamical regime, but with different expressive power. The small model produces proto-scientific fragments; the large model produces coherent expository paragraphs. The difference is one of **capacity**, not of **behaviour**: the underlying cognitive substrate is identical.

13.3 Thuvia, Maid of Mars

256 KB, 289 KB canonical, 21,722 unique samples, 404 sequences.

- Tiny (256 hidden, 32 output): 48K epochs, 2,575,439 transitions, 786 s, 6 MB model.
- Small (512 hidden, 64 output): 49K epochs, 2,634,890 transitions, 933 s, 17 MB model.
- Medium (1024 hidden, 128 output): 50K epochs, 2,696,169 transitions, 1466 s, 57 MB model.
- Large (2048 hidden, 256 output): 51K epochs, 2,737,131 transitions, 5946 s, 210 MB model.

Downscaling effects:

- longer output due to shallow attractors,
- fragmentary pulp-cadence,
- chaotic role binding,
- high creative drift.

Upscaling effects:

- longer narrative flow (6–20+ sentences),
- stable clause chains and syntactic inertia,
- strong Burroughs-style biome adaptation,
- reduced creative drift and increased canon rigidity.

13.4 General Observations

Across all datasets:

- GLPG's behaviour is reproducible: same dynamical regime, different capacity.
- Downscaled models produce longer outputs due to shallow attractor basins.
- Upscaled models compress output due to strong attractor geometry.
- Narrative coherence increases monotonically with hidden/output size.
- Creative drift decreases as attractor basins deepen.

GLPG therefore exhibits **cognitive scaling**: increasing capacity does not change the type of behaviour, but the *maturity* of the behaviour. Tiny models ramble; medium models narrate; large models stabilize into canon-rigid organisms.

Model	Tiny	Small	Medium	Large
Hidden / Output	256 / 32	512 / 64	1024 / 128	2048 / 256
Model Size MB	6	17	57	210
Output Length KB	147	122	112	104
Syntactic Stability	Low: telegraphic breaks	Medium: partial clause stability	High: consistent clause chains	Very high: inertial syntax
Narrative Flow	Micro-scenes, abrupt jumps	2–4 sentence continuity	6–12 sentence continuity	20+ sentence continuity
Role Binding	Weak, drifting roles	Moderate stability	Strong, consistent roles	Very strong, high inertia
Biome Adaptation	Shallow pulp vocabulary	Emerging pulp cadence	Strong Burroughs cadence	Fully internalized biome
Creative Drift	High, chaotic variation	Medium variation	Balanced creativity	Low, canon-locked
Semantic Coherence	Low, contradictions	Medium, partial logic	High, coherent scenes	Very high, rigid meaning
Recursion Robustness	Very low	Low–medium	High	Very high
Overall Cognitive Profile	*Infant attractor-core*	*Adolescent proto-organism*	*Adult cognitive substrate*	*Over-mature, canon-rigid organism*

Table 12: Comparison of four GLPG models trained on *Thuvia, Maid of Mars*. Scaling produces a clear cognitive development curve: tiny → small → medium → large.

14 GLPG as Cognitive Substrate

GLPG does not behave like a black-box language model. Across all ablations, GLPG responds to activation choice, hyperparameter variation, input-vector design, and recursion depth as a *dynamical system*. Its behaviour is logical, interpretable, and parametrizable. This makes GLPG fundamentally different from statistical transformer architectures: GLPG is a **cognitive substrate**.

14.1 Graph Optionality and Structural Orientation

In GLPG, grammar and lemma embeddings structurally orient the semantic manifold. As a result, the graph is optional: GLPG can operate in both graph-assisted and emergent-graph modes. In contrast, DMLG requires an explicit graph because its MLP cannot carry internal structure. This makes GLPG architecturally more elegant: the manifold itself contains the structural priors.

14.2 IEGR-Complete Cognitive Dynamics

GLPG is not a language model but a dynamic, strongly parametrizable, plastic, and modular cognitive substrate. Under hyperparameter variation, GLPG:

- deforms,
- stabilizes,

- evolves,
- and canonizes.

These logical behavioural shifts demonstrate that GLPG is an **IEGR-complete** architecture: it forms an internal emergent graph, a minimal but realistic demonstration of what an AGI substrate must be.

14.3 Cognitive Scale Laws

Biological and artificial cognitive systems exhibit the same pattern:

- too small: insufficient structure,
- optimal: maximal plasticity, variation, stability,
- too large: rigidity, redundancy, compression, telegraphic behaviour.

A cognitive substrate has an optimal size. Beyond that point, manifold compression reduces variation, narrows canon, and suppresses cognitive dynamics. GLPG shows this empirically, and similar scale laws will appear in transformer architectures once their internal graph representations saturate.

14.4 Cognitive Robustness Under Ablation

GLPG continues to function even when input, output, and MLP layers are heavily reduced. GLPG behaviour arises from *cognitive dynamics*, not from scale. This is precisely why GLPG is relevant: it is not a classical language model, but a cognitive substrate that emulates language through structural and semantic dynamics.

14.5 Style Adaptation and Canon Emergence

GLPG automatically adapts to writing style and canon biome without prompting or finetuning. Transformer models cannot do this without explicit conditioning. GLPG exhibits cognitive scale consistency and style emergence, not statistical correlation. It is a cognitive narrative system that adapts to any domain and produces emergent discourse forms.

14.6 Architecture vs. Dynamics

GLPG is a cognitive substrate that generates language. The architecture is constant; the dynamics scale. If language production were merely a statistical vector trick, humans would not be able to speak. Language requires a dynamic, plastic substrate with an internal energy envelope and a cognitive lifecycle.

14.7 Implications for AGI

AGI should be able to learn one book, not require a thousand books to reproduce one. AGI must be a dynamic, plastic IEGR substrate that functions at any scale, with a stable cognitive lifecycle.

GLPG demonstrates:

1. language generation without transformer mechanisms,
2. cognitive principles instead of statistical correlation,
3. semantic coherence emerging from vector dynamics,
4. full interpretability of internal mechanisms.

GLPG is therefore a minimal cognitive architecture that produces coherent language behaviour without attention mechanisms. This is a fundamental contribution to understanding how language can emerge from dynamical systems.

15 Architectural Innovations: Hybrid Recursion and Token History

GLPG's improved cognitive behaviour emerges from two architectural innovations: **Hybrid Recursion** in the forward pass and **Token History** in the input vector. Together, these mechanisms transform a simple MLP into a dynamic, directional, and semantically coherent cognitive substrate.

15.1 Hybrid Recursion: Internal Dynamical Refinement

The hybrid recursion block introduces controlled iterative refinement inside the forward pass. Instead of a single feedforward projection, GLPG performs multiple gated residual updates:

$$h \leftarrow h + g \cdot (u - h), \qquad g = \sigma(W_{\text{gate}} h), \qquad u = \text{silu}(W_{\text{refine}} h).$$

This mechanism increases the degree of recurrence (**R**) inside the substrate itself, independent of PRE (Proposal–Retry–Evaluate). Hybrid recursion provides:

- internal semantic refinement,
- attractor stabilization,
- clause-level coherence,
- controlled narrative drift,
- robustness under sampling variation.

Hybrid recursion is therefore a structural analogue of biological recurrent processing: a minimal form of cognitive iteration embedded directly in the substrate.

15.2 Token History: Sensory Continuity and Semantic Direction

Token History provides the substrate with a low-noise, lemma-driven temporal memory. It encodes:

- semantic continuity,
- temporal drift,
- decay-controlled retention,
- positional rotation,
- canon-consistent transitions.

Without Token History, GLPG remains functional but becomes **directionless**: syntax is coherent, but semantics collapse into telegraphic repetition. Without CSL (structural anchoring), Token History becomes **unstable**: semantic drift occurs without attractor control.

Only the combination **CSL + Token History** yields:

- coherent micro-scenes,
- stable narrative flow,
- selective semantic retention,
- controlled attractor geometry,
- emergent discourse structure.

This demonstrates the importance of **sensory input and memory** for any cognitive substrate: without structured perception, recurrence alone cannot produce meaningful behaviour.

15.3 Cognitive Synergy: Direction + Refinement

Hybrid recursion and Token History together form a minimal cognitive loop:

$$\text{Perception (Token History)} \rightarrow \text{Internal Refinement (Hybrid)} \rightarrow \text{Narrative Action (Output)}.$$

This loop gives GLPG:

- direction (semantic drift),
- stability (attractor basins),
- plasticity (controlled flux),
- coherence (recursive refinement),
- adaptability (biome and style emergence).

GLPG therefore behaves as a **cognitive substrate**, not a statistical LM: its behaviour arises from dynamical interaction between memory, structure, and recurrence.

15.4 Why These Innovations Matter

Hybrid recursion shows that recurrence inside the substrate increases cognitive capacity without increasing model size. Token History shows that meaningful cognition requires structured sensory input and temporal memory.

Together, they demonstrate:

- how semantic coherence can emerge from vector dynamics,
- how narrative flow can arise without attention mechanisms,
- how a minimal architecture can produce cognitive behaviour,
- how direction, stability, and emergence arise from simple principles.

GLPG is therefore not a transformer alternative, but a demonstration of how language can emerge from a dynamic system with recurrence, memory, and structural anchoring. It is a minimal cognitive architecture capable of producing coherent narrative behaviour.

16 Conclusion

This work introduced GLPG as a minimal yet expressive cognitive substrate capable of generating coherent narrative behaviour without transformer mechanisms. Across all ablations—activation functions, recursion depth, sampling parameters, input vector design, and cross-dataset evaluation—GLPG consistently behaved as a dynamical system rather than a statistical language model.

Two architectural innovations proved essential. The hybrid recursive forward pass provides internal semantic refinement and attractor stabilization, increasing the degree of recurrence inside the substrate itself. Token History supplies temporal continuity and directional semantic drift, demonstrating the importance of structured sensory input and memory for any cognitive system. Together, these mechanisms form a minimal cognitive loop that yields coherent micro-scenes, stable narrative flow, and emergent discourse structure.

Ablations showed that GLPG exhibits clear cognitive scale laws: too small leads to insufficient structure, optimal size yields maximal plasticity and stability, and too large produces rigidity and manifold compression. Unlike transformer models, which scale gradually with diminishing returns, GLPG undergoes categorical behavioural transitions. The minimum viable model and the full baseline share the same cognitive substrate, but differ in maturity: proto-narrative attractors versus coherent narrative flow.

Cross-dataset experiments demonstrated reproducibility. Whether trained on *Alice*, *Jekyll and Hyde*, *Outline of Science*, or *Thuvia, Maid of Mars*, GLPG maintained the same dynamical regime while adapting to each biome and stylistic canon. Scaling hidden and output dimensions produced predictable changes in cognitive behaviour, confirming that GLPG's architecture is stable across domains.

The results show that semantic coherence can emerge from vector dynamics, recurrence, and structured memory rather than attention mechanisms or large-scale statistical correlation. GLPG is not a transformer alternative but a demonstration of how language can arise from a compact, interpretable cognitive substrate. It remains functional even when heavily ablated, indicating that its behaviour is driven by cognitive principles rather than brute-force scale.

This work suggests that AGI substrates may not require massive corpora or attention-based architectures. A dynamic, plastic IEGR-complete system with structured input, temporal memory, and internal refinement may be sufficient to produce coherent language behaviour. GLPG provides a minimal, interpretable example of such a system, illustrating how narrative competence can emerge from simple but well-designed cognitive dynamics.

A Glossary of Cognitive and Architectural Terms

Attractor Basin
: A stable region of the semantic manifold toward which the model's internal state converges during generation. Deep attractor basins yield coherent, canon-consistent behaviour; shallow basins increase plasticity and creative drift.

Biome Adaptation
: The model's ability to internalise stylistic, lexical, and narrative patterns specific to a domain (e.g. Burroughs-style pulp cadence). Emerges from stable semantic manifolds rather than explicit conditioning.

Canon Pressure
: The tendency of a trained model to compress its narrative behaviour toward a stable internal world-model. Increases with model size, training depth, and output-embedding dimension.

Cognitive Substrate
: A dynamical system capable of producing coherent linguistic behaviour through structural and semantic dynamics rather than statistical correlation. GLPG is designed as such a substrate.

Cosine Scoring
: The similarity measure used to select the next lemma embedding in pure MLP mode. Determines how the semantic manifold is traversed during generation.

Flux
: The degree of semantic movement or dynamical energy in the manifold. Low flux produces rigid canon; high flux produces creative drift. Controlled by sampling parameters and activation functions.

Graph-Assisted Mode
: Generative regime in which deterministic last-lemma routing restricts candidate transitions. Provides structural cadence and collapse-resistance.

IEGR-Complete
: A system is IEGR-complete when it instantiates all four structural roles of intelligence—energetic efficiency (E), informational richness (I), geometric structure (G), and recursive depth (R)—in an

integrated and mutually reinforcing manner. As stated in the IEGR taxonomy, "Only systems that are both IEGR-complete and IEGR-scaled exhibit human-like general intelligence."

IEGR-completeness therefore requires:

- **E**: efficient cognitive transformation rather than brute-force scale,
- **I**: sufficiently rich and grounded semantic content,
- **G**: internal geometric organization, modularity, and canonical form,
- **R**: recursive self-refinement, multi-step reasoning, and reflective oversight.

A system lacking any of these roles falls into one of the parrot regimes described in the taxonomy (statistical, canonical, literal, or sophisticated parrots). IEGR-complete systems avoid these structural failure modes by integrating E, I, G, and R into a coherent cognitive architecture.

IEGR-completeness is a structural property, not an anthropomorphic one: it describes the minimal architecture required for general-purpose intelligence across biological, artificial, and hybrid substrates.

Manifold Width
The effective dimensional breadth of the semantic space. Too narrow yields telegraphic behaviour; too wide yields semantic diffusion. Optimal width depends on hidden-layer size and output-embedding dimension.

MLP Mode
Generative regime in which the full vocabulary is evaluated using cosine scoring. Exposes the learned semantic manifold and reveals attractor structure.

Narrative Flow
The degree to which generated text maintains coherent multi-sentence continuity. Strengthens with deeper attractor basins and stable semantic manifolds.

Paging Geometry
The number of candidate transitions per page in graph-assisted mode. Controls determinism, variation, and structural cadence.

Recursion Robustness
The ability of the model to remain coherent under repeated self-feeding generation. Increases with hybrid recursion and stable attractor basins.

Semantic Drift
Gradual movement of the model's internal semantic state across the manifold. Controlled by Token History, sampling parameters, and embedding blend coefficient.

Semantic Manifold
The learned geometry of grammar–lemma embeddings. Determines world-model depth, stylistic adaptation, and attractor structure.

Structural Anchoring
The grounding provided by CSL (Current + Sequence + Line). Prevents semantic drift from collapsing into repetition and ensures clause-level coherence.

Token History
A lemma-driven, decay-controlled temporal memory that provides semantic continuity and directional drift. Essential for coherent narrative behaviour.

Transformative Phase Change
A categorical shift in cognitive behaviour caused by changes in model capacity, rather than a gradual improvement. GLPG exhibits such transitions between MVM, baseline, and large models.

B Triadic Cosmos Ecosystem Context

The present paper is part of the broader *Triadic Cosmos* ecosystem: a unified research program that develops a compositional ontology of intelligence together with modular, inspectable computational architectures. The ecosystem spans conceptual theory, executable reference implementations, and fully documented empirical demonstrations, forming a coherent framework for studying hybrid intelligent systems.

All materials, extended notes, and evolving documentation are publicly available in the open GitHub project:

`https://github.com/triadic-cosmos`

Within the `triadic-system` repository, a complete reference implementation of the GLPG engine, the narrator pipeline, and all supporting modules is provided under the AGPL-3.0 license. The datasets described in this paper are present within the complementary `triadic-data` repository.

Closing Note. This paper does not present GLPG as an artificial general intelligence system. Its purpose is to investigate GLPG as a compact, interpretable cognitive substrate that enables controlled, reproducible experimentation on AGI-relevant dynamics. Within the Triadic Cosmos ecosystem—and equally outside it—GLPG serves as a minimal testbed for studying attractor formation, semantic manifold development, structural scaling laws, and cognitive phase transitions.
The broader ecosystem research program does not aim to construct AGI, but to clarify and empirically explore the structural conditions under which general intelligence could emerge. GLPG is therefore a scientific instrument: a substrate for AGI research, not an AGI system.

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